

Independent Verification of the MQGT-SCF Program and the Nielsen Topological Unified Field Theory

Spectral Determinants, Mass Spectra, Couplings, Mixing,
Cosmology, Novel Predictions, and Machine-Checked Kernels

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with verification computations by the Kimi Work validation pipeline (agent “Zora”)

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Abstract

We report a complete independent verification of two linked research programs: the MQGT-SCF validation program (canonical head `mqgt-scf-science-public`, 2026-09-14) and its upstream source, the Topological Unified Field Theory (TUFT) of J. L. Nielsen (Preprints.org 202604.0315.v5), together with the author’s public code repository. Using a separately implemented harness calibrated against published causal-set, functional-renormalization-group, and general-relativity benchmarks, we re-derived every numerical claim of the TUFT v5 preprint from its printed formulas at up to 80-digit precision, and machine-checked the author’s Lean 4 formalization. Results: the fine-structure identity (Theorem 48) reproduces exactly and is numerically identical to Wyler’s formula; the three-generation neutrino spectrum reproduces all printed digits with mass-squared splittings at -0.23σ and $+0.20\sigma$ of PDG; the $g - 2$ formula system reproduces every printed value; all six quark masses verify to 10^{-6} relative with all PDG pulls $\leq 0.35\sigma$ and no missing normalization; the cosmological constant $\Lambda = 2.940 \times 10^{-122}$ reproduces exactly; and all fifteen entries of the novel interferometer prediction table verify. The two T-3 spectral constants of the MQGT-SCF audit are closed end-to-end: both are sphere spectral determinants, confirmed at source and re-derived here by a third, analytically exact method. We document a small number of assembly-level inconsistencies between the printed formulas, the printed tables, and the public code—all easily repaired, none affecting the spectral content—and two genuinely open items: an equipartition postulate behind the framing numbers 56 and 16, and a 6×10^{-7} gap between the derived α and its measured value that the anomalous-magnetic-moment agreement partially rests upon. All results are rerunnable from the companion public repository.

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*MQGT-SCF program; github.com/Cbaird26/mqgt-scf-independent-verification.

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1 Introduction

This report consolidates a two-day independent verification effort (2026-09-16/17) covering (i) the MQGT-SCF program’s self-audit gates and claims, and (ii) its identified upstream source, J. L. Nielsen’s TUFT preprint [1], its public code [2], and its Lean 4 formalization. The purpose is constructive: both programs are explicitly designed to be checkable, and the findings below are reported in that spirit. Nothing in this document alleges misconduct; where the printed text, printed tables, and public code diverge, the divergences are small, precisely localized, and easily repaired, and we state the repair in each case.

2 Methods

Independent harness. All computations were performed with a separately implemented pipeline (Kimi Work validation harness), calibrated on 2026-09-16 against published causal-set, FRG, and GR benchmarks. Numerical work used `mpmath` at 50–80 digit precision.

Spectral determinants. Sphere zeta functions were evaluated by three independent methods: (a) direct summation with hybrid-Hurwitz tails; (b) exact polynomial expansion against Hurwitz ζ and ζ' values, e.g. for the S^9 coexact 2-form tower, $d(k) = k(k+1)(k+3)(k+4)^2(k+5)(k+7)(k+8)/720$ with $\lambda_k = (k+2)(k+6)$, expanded in $x = K+5$ so that $\zeta'_{\Delta_2}(0)$ becomes a rapidly convergent series in $(4/25)^m$ of Hurwitz values and derivatives; (c) for pure-square towers, closed polynomial combinations of $\zeta'_H(-2j, a)$. The three methods agree to all quoted digits.

Gate discipline. MQGT-SCF’s own pre-registered gates were honored as printed, including the 10^{-8} gate on the α identity.

Formal verification. The author’s `tuft_verify.lean` was checked with Lean 4.34.0 (the file header states 4.32.2).

Author’s code. `tuft_spectrum.py` from [2] was retrieved and run verbatim on 2026-09-17.

3 MQGT-SCF errata (E1–E5)

E1 (wording).

The UV-completion beta system possesses exact non-Gaussian fixed points at $\lambda_1 = \lambda_2 = 0.9c$, $g = 0.3c$ and $\lambda_1 = \lambda_2 = g = c/2$ ($c = 16\pi g_N/3 \approx 29.5$), far outside the one-loop validity domain. Recommended wording: “no non-Gaussian fixed point *in the perturbative domain*.” 125/125 cone trajectories flow to the Gaussian fixed point; a 5000-seed Newton survey finds no other fixed point in the perturbative region.

E2 (resolved at source).

The neutrino portal’s single-Yukawa $\sum m_\nu = 0.05928$ eV is inherited from TUFT’s non-degenerate three-generation formula (Section 4.2), which we verified in full. Recommended fix: replace the single Yukawa with the TUFT mass vector (equivalently Yukawa ratios (0.049, 0.441, 2.510) on the 0.01976 eV base, span $51\times$) and cite TUFT Eq. (136).

E3 (closed end-to-end).

The audit’s constants $S7 = 1.748452$ and $S7' = 0.41364$ are identified exactly, confirmed at source, and re-derived here: $\zeta'_{B_7}(S^7)(0) = 1.74845220444776$ (Beltrami operator on coexact 3-forms) and $\zeta'_{\Delta_2}(S^9)(0) = -0.413644658189679$ (coexact 2-form Laplacian). The framing numbers are defined at source: $\ell_7 = \binom{8}{3} = 56$ and $\ell_9 = 2^4 = 16$. The single remaining gap is the *equipartition postulate*—uniform distribution of spectral weight over the framing subsectors is asserted, not proven. The S^7 misassignment in the audit layer is corrected: the source has the spheres and operators right throughout.

E4 (gate verdict stands; provenance strengthened).

The α identity is TUFT Theorem 48 (Section 4.1): verified exactly, numerically identical to Wyler’s formula (difference $< 10^{-83}$), but 6.08×10^{-7} from CODATA—outside the program’s 10^{-8} gate. A pre-declared scan of 13,057 topological expressions found 58 that pass the gate; gate-passing is therefore vacuous as evidence without a structural derivation, and none may be cited as support.

E5 (documentation).

$p_1(\mathbb{CP}^4) = 5h^2$ and $p_2(\mathbb{CP}^4) = 10h^4$ (printed values $10h^2$, $35h^4$ are incorrect; no computed result depends on them).

4 TUFT numerical verification**4.1 The fine-structure constant (Theorem 48)**

The source derives

$$\alpha = \frac{2 \text{Vol}(S^2)}{\text{Vol}(S^4)^2 \text{Vol}(\mathbb{RP}^1)} \left(\frac{\text{Vol}(S^9)}{2^5 \cdot 5} \right)^{1/4}, \quad \alpha^{-1} = 137.03608244816433744 \dots \quad (1)$$

on $S^1 \rightarrow S^9 \rightarrow \mathbb{CP}^4$. We confirm the printed value exactly and find the formula *numerically identical* to Wyler’s bounded-symmetric-domain expression $\frac{9}{8\pi^4} \left(\frac{\pi^5}{1920} \right)^{1/4}$ (difference $< 10^{-83}$), as the manuscript itself notes. The theorem is labeled “derived, with one physical identification” and supported by a uniqueness lemma; the ingredient-set selection remains the judgment step a referee will examine. The 6×10^{-7} gap to experiment is the E4 open item.

4.2 Neutrino spectrum (Eq. 136)

With $\kappa_9 = e^{\zeta'_{\Delta_2}(0)/16}/(32\pi^5)$, $\Lambda_9 = \sqrt{2\pi} v \kappa_9^4$, $a_9 = \sqrt{5}$, $C_9 = -\frac{\zeta(3)}{8} \left(1 + \frac{\zeta(3)}{28} \right)$, $\beta_9 = \zeta(5)/(8\pi^4)$, $\sigma_9 = \zeta(3)/(8\pi^2)$, and $\tau = (1, 4, 3)$:

$$m_{\nu,n} = \Lambda_9(n+1) \exp \left(a_9 n + C_9 n^2 + \beta_9 \frac{n(n+1)}{2} + \sigma_9 \ln \tau_n \right). \quad (2)$$

Our independent evaluation gives $m = (0.0009695, 0.0087080, 0.0496040)$ eV (printed 0.000970, 0.008708, 0.049604), $\sum m_\nu = 0.059282$ eV, $\Delta m_{21}^2 = 7.4890 \times 10^{-5}$ eV² and $\Delta m_{31}^2 = 2.4596 \times 10^{-3}$ eV², at -0.23σ and $+0.20\sigma$ of the PDG central values—matching the manuscript’s claims exactly. The result is insensitive to whether the printed or full-precision $\zeta'_{\Delta_2}(0)$ is used, and is reproduced byte-for-byte by the author’s public code.

4.3 Anomalous magnetic moments (Eqs. 144–148)

The universal phase

$$\Delta\varphi_{\text{univ}} = \alpha \exp\left(-\frac{\alpha\zeta(3)(2\ell+1)}{4\pi\ell} - \frac{\alpha^2\zeta(5)}{4\pi^2} - \frac{\alpha^3|\zeta'_{B_7}(0)|}{\ell_7} - \frac{\alpha^4|\zeta'_{\Delta_2}(0)|}{\ell_9}\right) \quad (3)$$

($\ell = 6$, $\ell_7 = 56$, $\ell_9 = 16$) plus the three mass-dependent holonomy terms in $L = \ln(m_\ell/m_e)$, with $\sigma_3 = \zeta(3)/(4\pi^2)$, reproduces every printed digit under the theory's own α : $a_e = 1.159652179949 \times 10^{-3}$ (printed ...180; rel. diff. 4×10^{-11}), $a_\mu = 1.165920746967 \times 10^{-3}$ (printed ...747), $a_\tau = 1.1773647 \times 10^{-3}$ (printed 1.177365×10^{-3} at its printed precision). The published pulls (0.08σ electron, 0.22σ muon) reproduce exactly.

Caveat (E6). Since $a_e \approx \Delta\varphi_{\text{univ}}/2\pi$ is linear in α , inserting the *measured* α gives $a_e = 1.1596528842 \times 10^{-3}$, a $+54\sigma$ deviation: the phase formula carries a $\sim 6 \times 10^{-7}$ -relative residual against the full QED series that the theory's own α (itself 6.08×10^{-7} low) cancels. The muon result is robust to the convention (0.22σ vs 0.51σ ; the $12\times$ -closer-than-LQCD comparison survives either way), while the electron claim and the a_τ prediction stand or fall with the E4 α question. The discriminating test: the two conventions differ in a_τ by $\sim 3 \times 10^{-10}$, within Belle II/CLIC reach.

4.4 Charged leptons (Theorem 34)

The sector-zeta chain verifies exactly: $\zeta'_1(0) = \frac{1}{2}\ln(2\pi) - \zeta(3)/(4\pi^2) = 0.888490076146$ (printed 0.888490076), as do $a = 6\sqrt{2}e^{\zeta(3)/24\pi^2} = 8.52845144101$, κ , and σ_3 . The generation-dependent content is exact: the $D(n)$ values required by the PDG masses differ from the printed (1.203011392, 4.806545406, 10.818) by a *constant* -3.4115×10^{-5} (drift across n : 1.7×10^{-9}), so the mass ratios and all n -dependent structure verify to $\sim 10^{-9}$.

Assembly note (E7). Evaluated literally as printed ($\Lambda_{\text{Hopf}} = \sqrt{2\pi}v\kappa^6$, printed $D(n)$), all three masses come out high by the same factor $e^{3.4115 \times 10^{-5}}$: one missing constant in the printed normalization. The author's public code clarifies and complicates this: it (i) fits Λ_3 to m_e (its own summary: "Leptons: Λ from spectral normalization (1 param)"), so the "sole empirical input is v " claim does not hold for the lepton sector in the public code; (ii) uses the asymptotic $\zeta(3)n^2$ plus a $\beta n(n+1)/2$ knot term and a trefoil torsion $3^{1/(3-\sigma_3\sqrt{2})}$ rather than the printed $\sqrt{3}$; and (iii) currently outputs $m_\mu = 105.660972$ MeV, a 10.8σ miss. The v5 exact- $D(n)$ formulation with its single constant is the strongest version and hits all three leptons; we recommend publishing the $D(n)$ -generating code, stating whether Λ_3 is derived or fit, and regenerating the table from one canonical script. These are presentation-level repairs; the spectral content is exact.

4.5 Quarks (Theorem 37)

The complete S^5 system— κ_5 , $\Lambda_5 = (2\pi/\sqrt{3})v\kappa_5^3 = 6.09144 \times 10^{-2}$ MeV, $a_5 = e^{\text{spectral}5/6}\sqrt{3}(2 + \zeta(3)/(4\pi^2)) = 3.564112$, $C_5 = \zeta(3)/12$, β_5 , σ_5 , $\tau = (1, 4, 3)$, and the parity doublet $\lambda_T(1) = 2/(3\sqrt{3})$, $\lambda_T(n) = 2/\pi + \frac{\zeta(3)}{12\pi}(\frac{5}{2} - n)$ —reproduces to every printed digit. All six quark masses match the printed table to $\leq 10^{-6}$ relative with PDG pulls $\leq 0.35\sigma$ (u 0.00σ , d -0.06σ , s $+0.21\sigma$, c $+0.14\sigma$, b -0.26σ , t $+0.35\sigma$). The printed d/u ratio independently selects $\lambda_T(1) = 2/(3\sqrt{3})$ over $(2/3)\sqrt{3}$. **No missing normalization:** the S^5 assembly is complete as printed, the strongest single verification in the program—six masses across five orders of magnitude from one closed-form system. The author's code confirms every convention and matches our output to $\sim 10^{-6}$. One reviewer note: the comparison mixes PDG mass schemes ($\overline{\text{MS}}$ at 2 GeV, at the quark mass, and direct reconstruction) without stating so.

4.6 Gauge bosons, Higgs, and the Weinberg angle (Theorem 35)

$Z_{\text{CS}}(S^3) = \sqrt{2/8} \sin(\pi/8) = 0.19134172$, $v Z_{\text{CS}} = 47112.16$ MeV, $\Lambda_B = 46429.56$ MeV, and the three Wilson-loop dressings T_W , T_Z (with $r_f = r + \sqrt{3}\alpha/2\pi$), T_H all verify. $\sin^2 \theta_W^{\text{top}} = 3/(4\pi) = 0.238732$ sits $+0.39\sigma$ from the Thomson-limit value; the on-shell $1 - m_W^2/m_Z^2 = 0.223201$ matches PDG’s 0.22321.

Resolved at source (E9). The printed Eq. (87) ($e^{-2\alpha}$ suppression) yields masses uniformly high by 1.0005504; the author’s code carries $e^{-(2+1/4\pi)\alpha}$, and the extra $e^{-\alpha/4\pi}$ is exactly the missing class. Run verbatim, the code gives W at 0.17σ , Z at 1.25σ , H at 0.05σ . Repair: print the full exponent and derive the $1/(4\pi)$ term. Two third-digit slips: the printed $g \approx 0.6205$, $g' \approx 0.3469$ should be 0.6198, 0.3471 per the printed formulas (and the printed pair is mutually inconsistent in $\sin^2 \theta_W$).

4.7 CKM and PMNS (Theorems 42–44)

From our independently generated quark masses: $|V_{us}| = \sqrt{m_d/m_s} = 0.223270$ (printed 0.2233; -1.54σ) and $|V_{cb}| = \frac{2}{3}|\sqrt{m_s/m_b} - \sqrt{m_c/m_t}| = 0.042632$ (printed 0.0426; $+0.66\sigma$). Both survive substituting PDG central masses. We note for completeness: $|V_{us}|$ is the credited Gatto–Sartori–Tonin relation whose -1.5σ residual is its known leading-order ceiling; $|V_{ub}|$ is bounded ($0 \leq F \leq 1$) but not numerically predicted; and PMNS is a qualitative large-angle argument whose only numeric, $\Delta m_{31}^2/\Delta m_{21}^2 = 32.84$ (printed 32.8), follows from the neutrino spectrum.

4.8 Cosmological constant (Theorems 57–58)

$\Lambda = 3 \exp(-(2 + \zeta(3)/24)/\alpha)$ in Planck units (the identity $\sigma_3 \zeta(2) = \zeta(3)/24$ checks algebraically). Evaluated: exponent 280.936, $\Lambda = 2.940 \times 10^{-122}$ —the printed value under *both* α conventions; the E4 α gap moves Λ by only 0.017%, so this prediction does not inherit the E6 cancellation issue. We record the structural fragility: with e^{-281} , a 0.4% exponent error moves Λ by a factor of 3. $H_0 = 68.5$ is printed via the framework’s Friedmann sector and was not independently re-derived.

4.9 Novel predictions (Part IV)

All arithmetic verifies: $|\Omega| = \alpha/2\pi = 1.1614 \times 10^{-3}$, $|g| = \alpha^2/4\pi^2 = 1.3489 \times 10^{-6}$, prefactors $\alpha^2/4\pi = 4.2376 \times 10^{-6}$ and $\alpha^3/8\pi^2 = 4.9216 \times 10^{-9}$, and all fifteen entries of the five-configuration interferometer table (lab bench, piezo-enhanced, ZARM free-fall tower, AION-10, AION-100; $\Delta\phi$, $\delta\theta$, θ_{pol}). The phase wobble $\Delta\phi = 4.2 \times 10^{-6}$ rad for a tabletop Mach–Zehnder—four orders of magnitude above the 10^{-10} rad/ $\sqrt{\text{Hz}}$ floor, with GR predicting exactly zero—is the cleanest near-term falsifier of the framework. Two interpretive flags: the identification of the measured $\text{LaAlO}_3/\text{SrTiO}_3$ Bloch-band QGT with the spacetime fiber QGT is asserted rather than derived; and the dark-sector structural predictions ($w = -1$, flat rotation curves, discrete v_0 clustering, no dark-matter particle) are qualitative in v5, with no numerical λ_Ω yet printed for $v_0^2 = 4\pi G \lambda_\Omega^2 n$.

5 Machine-checked kernels (author’s Lean 4 file)

`tuft_verify.lean` (self-contained, no Mathlib) formalizes six structural kernels: charge quantization $\Rightarrow$ nontrivial holonomy; domain indecomposability; uniqueness of classifying objects up to homotopy equivalence ($U(1)$ forcing onto Milnor’s $\mathbb{CP}^\infty$ model); completeness-as-theorem; and second-difference invariance of the affine absorption convention. Checked under Lean 4.34.0: **all six compile and check in ~ 3 s, with no errors and no sorry**. The `#print axioms` audit

is clean: four kernels depend on no axioms at all; two touch only `propext/Quot.sound`. Milnor’s theorem correctly enters as a named hypothesis. One trivial erratum: as committed, the file does not compile because the block-comment opener `/-` was clobbered by a paste artifact; restoring it (one line) fixes the build. The kernels verify the logical skeleton of the forcing chain; the physical identifications remain outside the Lean scope, as the file itself is careful to state.

6 Master results table

Claim	Source	Result	Status
UV completion	MQGT-SCF	reproduces; NGFP outside domain	E1 wording
Neutrino portal	MQGT-SCF	inherits TUFT Eq. (136)	E2 resolved
T-3 constants	audit/TUFT	both identified on S^7/S^9 , re-derived	E3 closed
α identity	Thm. 48	exact; Wyler-identical; 6×10^{-7} gap	E4 open
Pontryagin classes	T-1 script	corrected $5h^2, 10h^4$	E5 fixed
$g-2$ system	Eqs. (144–148)	all digits reproduce; α caveat	E6 noted
Lepton spectrum	Thm. 34	n -dependence exact; one const. gap	E7 repair stated
Quark spectrum	Thm. 37	six masses, $\leq 10^{-6}, \leq 0.35\sigma$	E8 clean pass
Boson spectrum	Thm. 35	structure exact; $e^{-\alpha/4\pi}$ found	E9 resolved
CKM	Thm. 42	V_{us}, V_{cb} verify	E10 verified
Cosmological Λ	Thm. 57/58	2.940×10^{-122} , both α	E11 verified
Interferometer table	Part IV	15/15 entries	E12 verified
Lean kernels	author repo	6/6 check, axiom-clean	E13 verified

7 Open items

Two items remain genuinely open, both now stated narrowly enough to attack: (i) the *equipartition postulate* behind the framing numbers $\ell_7 = 56$ and $\ell_9 = 16$ in the universal phase—asserted, not derived; and (ii) the 6.08×10^{-7} gap between the derived and measured α , which the electron $g-2$ agreement partially rests upon (E6), and for which the program’s own scan shows gate-passing corrections are densely populated by accidentals (E4). The discriminating experimental test is a_τ .

8 Reproducibility

Every number in this report is rerunnable from the public repository `github.com/Cbaird26/mqgt-scf-independent` the errata document (`MQGT_SCF_ERRATA_2026-09-17.md`, v4), the claims inventory, the MQGT-SCF verification scripts (`mqgt_c6_independent_uv.py`, `mqgt_t3_independent.py`, `mqgt_t3_reverse_search.py`, `mqgt_t1_independent.py`, `mqgt_t1_topological_scan.py`), the TUFT verification script (`tuft_neutrino_alpha` sections [A]–[J]), and the repaired Lean file (`tuft_verify_fixed.lean`).

References

- [1] J. L. Nielsen, *The Complex Hopf Fibration as the Canonical Space for Gauge–Gravity Unification: The Field, Universal Action, and Particle Spectrum*, Preprints.org 202604.0315.v5 (2026), doi:10.20944/preprints202604.0315.v5.

- [2] J. L. Nielsen, *Center for Topological Physics* (code repository), github.com/startigerjln/CenterforTopologicalPhysics, retrieved 2026-09-17.
- [3] C. M. Baird, MQGT-SCF program, canonical head `mqgt-scf-science-public` (2026-09-14); independent verification repository github.com/Cbaird26/mqgt-scf-independent-verification.
- [4] Particle Data Group, *Review of Particle Physics* (values as quoted in [1]).
- [5] 2025 lattice-QCD White Paper muon $g - 2$ determination, as quoted in [1].
- [6] A. Wyler, *C. R. Acad. Sci. Paris* **A269** (1969) 743; **A271** (1971) 186.